

Status Register

Test Report

Comprehensive Verification of PolyFlop8 Status Register

Document ID:	TC-SREG-001
Revision:	1.0
Date:	January 5, 2026
Test Level:	Unit/Block Level
Test Engineer:	Artem Horiunov
Status:	PASS

1 TEST OVERVIEW

1.1 Unit Under Test (UUT)

Component	Status Register for PolyFlop8 Processor
VHDL Entity	StatusReg
Source File	StatusReg.vhd, tb_StatusReg.vhd
Test Environment	ModelSim/QuestaSim 2023.4, VHDL Testbench
Register Width	8-bit

1.2 Scope

This test report documents the verification of the Status Register for the PolyFlop8 Processor. The Status Register is responsible for storing and managing the processor's status flags, including interrupt enable, sign, overflow, negative, zero, and carry flags.

1.3 Test Objectives

- Verify correct operation of the Status Register during reset
- Validate write and read functionality for all flags
- Confirm correct calculation of the S-Flag (Sign Flag)
- Test preservation of I and T bits during ALU updates
- Ensure 100% code coverage of Status Register behavioral description

2 REQUIREMENTS COVERAGE

Req ID	Requirement Description	Test Case ID	Status
TR-SREG-01	Verify ALU Update	TC-SREG-001-01/04	PASS
TR-SREG-02	Verify Restore Mode	TC-SREG-001-02	PASS
TR-SREG-03	Verify S-Flag Calculation	TC-SREG-001-03	PASS
TR-SREG-04	Verify I/T Bits Preservation	TC-SREG-001-01/04	PASS
TR-SREG-05	Verify Reset Functionality	TC-SREG-001-05	PASS

Table 1: Requirements Coverage Matrix

3 DETAILED TEST CASES

3.1 Test Case TC-SREG-001-05: Reset Verification

Requirement: TR-SREG-05
Purpose: Verify reset functionality and initial state.
Status: PASS

3.2 Test Case TC-SREG-001-02: Restore Mode (Load 0xAA)

Requirement: TR-SREG-02

Purpose: Verify restore mode functionality by loading 0xAA.

Status: PASS

3.3 Test Case TC-SREG-001-01/04: ALU Update (Preserve I/T bits)

Requirement: TR-SREG-01, TR-SREG-04

Purpose: Verify ALU update functionality and preservation of I/T bits.

Status: PASS

3.4 Test Case TC-SREG-001-03: S-Flag Calculation (N=1, V=0 -> S=1)

Requirement: TR-SREG-03

Purpose: Verify S-Flag calculation with N=1 and V=0.

Status: PASS

3.5 Test Case TC-SREG-001-03: S-Flag Calculation (N=1, V=1 -> S=0)

Requirement: TR-SREG-03

Purpose: Verify S-Flag calculation with N=1 and V=1.

Status: PASS

4 SOURCE CODES

4.1 StatusReg.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity StatusReg is
    Port (
        clk          : in  STD_LOGIC;
        rst          : in  STD_LOGIC;
        sreg_we       : in  STD_LOGIC;
        sreg_src      : in  STD_LOGIC;
        alu_flags     : in  STD_LOGIC_VECTOR (4 downto 0);
        regfile_data_b : in  STD_LOGIC_VECTOR (7 downto 0);
        sreg_out      : out STD_LOGIC_VECTOR (7 downto 0)
    );
end StatusReg;

architecture Behavioral of StatusReg is
    signal sreg : std_logic_vector(7 downto 0);
begin
    process(clk, rst)
    begin
        if rst = '1' then
            sreg <= (others => '0');

        elsif rising_edge(clk) then
            if sreg_we = '1' then
                if sreg_src = '1' then
                    sreg <= regfile_data_b;
                else
                    sreg(0) <= alu_flags(1);
                end if;
            end if;
        end if;
    end process;
end Behavioral;
```

```

        sreg(1) <= alu_flags(0);
        sreg(2) <= alu_flags(2);
        sreg(3) <= alu_flags(3);
        sreg(4) <= alu_flags(2) xor alu_flags(3);
        sreg(5) <= alu_flags(4);
        sreg(6) <= sreg(6);
        sreg(7) <= sreg(7);
    end if;
end if;
end if;
end process;

sreg_out <= sreg;

end Behavioral;

```

4.2 tb_StatusReg.vhd

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity tb_StatusReg is
end tb_StatusReg;

architecture Behavioral of tb_StatusReg is

    component StatusReg
        Port (
            clk            : in  STD_LOGIC;
            rst            : in  STD_LOGIC;
            sreg_we        : in  STD_LOGIC;
            sreg_src        : in  STD_LOGIC;
            alu_flags       : in  STD_LOGIC_VECTOR (4 downto 0);
            regfile_data_b  : in  STD_LOGIC_VECTOR (7 downto 0);
            sreg_out        : out STD_LOGIC_VECTOR (7 downto 0)
        );
    end component;

    signal clk_tb          : STD_LOGIC := '0';
    signal rst_tb          : STD_LOGIC := '0';
    signal sreg_we_tb      : STD_LOGIC := '0';
    signal sreg_src_tb     : STD_LOGIC := '0';
    signal alu_flags_tb    : STD_LOGIC_VECTOR(4 downto 0) := (others => '0');
    signal regfile_data_b_tb : STD_LOGIC_VECTOR(7 downto 0) := (others => '0');
    signal sreg_out_tb     : STD_LOGIC_VECTOR(7 downto 0);

    constant CLK_PERIOD : time := 10 ns;

    function to_hex_string(slv : std_logic_vector) return string is
    begin
        return integer'image(to_integer(unsigned(slv)));
    end function;

begin

    uut: StatusReg port map (
        clk      => clk_tb,
        rst      => rst_tb,
        sreg_we   => sreg_we_tb,
        sreg_src  => sreg_src_tb,
        alu_flags => alu_flags_tb,

```

```

    regfile_data_b => regfile_data_b_tb,
    sreg_out       => sreg_out_tb
);

clk_process: process
begin
    clk_tb <= '0';
    wait for CLK_PERIOD/2;
    clk_tb <= '1';
    wait for CLK_PERIOD/2;
end process;

stim_proc: process
begin
    report "TC-SREG-05: Reset Verification";
    rst_tb <= '1';
    wait for CLK_PERIOD * 2;
    rst_tb <= '0';
    wait for CLK_PERIOD;
    assert sreg_out_tb = x"00"
        report "Error TC-SREG-05: Reset failed. Expected 0x00, got " &
            ↳ to_hex_string(sreg_out_tb)
        severity error;

    report "TC-SREG-02: Restore Mode (Load 0xAA)";
    sreg_src_tb <= '1';
    regfile_data_b_tb <= x"AA";
    sreg_we_tb <= '1';
    wait for CLK_PERIOD;
    sreg_we_tb <= '0';
    wait for 1 ns;
    assert sreg_out_tb = x"AA"
        report "Error TC-SREG-02: Restore failed. Expected 0xAA, got " &
            ↳ to_hex_string(sreg_out_tb)
        severity error;

    report "TC-SREG-01/04: ALU Update (Preserve I/T bits)";
    sreg_src_tb <= '0';
    alu_flags_tb <= "00000";
    sreg_we_tb <= '1';
    wait for CLK_PERIOD;
    sreg_we_tb <= '0';
    wait for 1 ns;
    assert sreg_out_tb = x"80"
        report "Error TC-SREG-04: I-bit preservation failed. Expected 0x80, got " &
            ↳ to_hex_string(sreg_out_tb)
        severity error;

    report "TC-SREG-03: S-Flag Calculation (N=1, V=0 -> S=1)";
    alu_flags_tb <= "00100";
    sreg_src_tb <= '0';
    sreg_we_tb <= '1';
    wait for CLK_PERIOD;
    assert sreg_out_tb = x"94"
        report "Error TC-SREG-03 (Case 1): S-flag failed. Expected 0x94, got " &
            ↳ to_hex_string(sreg_out_tb)
        severity error;

    report "TC-SREG-03: S-Flag Calculation (N=1, V=1 -> S=0)";
    alu_flags_tb <= "01100";
    wait for CLK_PERIOD;
    sreg_we_tb <= '0';
    assert sreg_out_tb = x"8C"

```

```

        report "Error_TC-SREG-03(Case_2): S-flag failed. Expected 0x8C, got
        ↪ " & to_hex_string(sreg_out_tb)
        severity error;

        report "---StatusRegisterTestsCompletedSuccessfully---";
        wait;
    end process;

end Behavioral;

```

5 SIMULATION LOG

```

Time resolution is 1 ps
Note: TC-SREG-05: Reset Verification
Time: 0 ps Iteration: 0 Process: /tb_StatusReg/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ StatusRegister_TB.vhd
Note: TC-SREG-02: Restore Mode (Load 0xAA)
Time: 30 ns Iteration: 0 Process: /tb_StatusReg/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ StatusRegister_TB.vhd
Note: TC-SREG-01/04: ALU Update (Preserve I/T bits)
Time: 41 ns Iteration: 0 Process: /tb_StatusReg/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ StatusRegister_TB.vhd
Note: TC-SREG-03: S-Flag Calculation (N=1, V=0 -> S=1)
Time: 52 ns Iteration: 0 Process: /tb_StatusReg/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ StatusRegister_TB.vhd
Note: TC-SREG-03: S-Flag Calculation (N=1, V=1 -> S=0)
Time: 62 ns Iteration: 0 Process: /tb_StatusReg/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ StatusRegister_TB.vhd
Note: --- Status Register Tests Completed Successfully ---
Time: 72 ns Iteration: 0 Process: /tb_StatusReg/stim_proc File: F:/Projects/
    ↪ PolyFlop8-MCU/Polyflop8/Polyflop8.srscs/sources_1/new/Unit_test_bench/
    ↪ StatusRegister_TB.vhd

```

6 PASS/FAIL CRITERIA

6.1 Success Criteria

Criteria	Description
Functional	All test cases execute without assertion failures
Timing	No timing violations (Status Register is synchronous)
Coverage	100% statement coverage, 100% branch coverage achieved
Consistency	Identical results across multiple simulation runs
Documentation	All test results properly documented and signed off

6.2 Failure Conditions

- Any single assertion failure during simulation

- Code coverage less than 100% for Status Register behavioral code
- Timing violations or simulation warnings
- Inconsistent results between simulation runs
- Missing or incomplete test documentation

6.3 Coverage Goals

- **Statement Coverage:** 100% of Status Register process statements executed
- **Branch Coverage:** 100% of case statement branches taken
- **Condition Coverage:** All flag calculation conditions tested
- **Toggle Coverage:** All signal bits toggled 0→1 and 1→0
- **FSM Coverage:** N/A (Status Register is not a state machine)

7 SPECIAL TEST CONSIDERATIONS

7.1 Edge Cases

- Test all flag combinations with minimum and maximum values
- Verify proper handling of S-Flag calculation
- Test boundary conditions for flag inputs
- Verify correct behavior during clock edge transitions
- Test all possible multiplexer combinations

7.2 Timing Considerations

- Status Register is synchronous - verify setup and hold times
- Monitor for glitches or hazards during input transitions
- Verify stable outputs within specified propagation delay
- Test with minimum, typical, and maximum delay models

7.3 Integration Considerations

- Verify interface compatibility with ALU and Control Unit
- Test handshake signals with Control Unit
- Validate output format matches processor expectations
- Confirm timing compatibility with other processor components